

● EASY

● ALGEBRA

Mixed

10 questions · ~15 min

04

Q1

ALGEBRA

$$x + y = 18$$

$$5y = x$$

What is the solution x, y to the given system of equations?

- A. 15, 3
- B. 16, 2
- C. 17, 1
- D. 18, 0

Q2

ALGEBRA

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

- A. 30
- B. 22
- C. 11
- D. -23

Q3

ALGEBRA

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $fx = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of 33 in this context?

- A. The estimated height, in cm, of the liquid at the start of the project
- B. The estimated height, in cm, of the liquid at the end of the project
- C. The estimated change in the height, in cm, of the liquid each day
- D. The estimated number of days for all of the liquid to evaporate

Q4

ALGEBRA

To repair a refrigerator, a technician charges \$ 60 per hour for labor plus \$ 120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

- A. $fx = x + 120$
- B. $fx = 60x$
- C. $fx = 60x + 120$
- D. $fx = 60x - 120$

A group of monarch butterflies migrated from Chicago, Illinois, to Michoacán, Mexico, flying a total of 2,100 miles. It took a single butterfly in the group 120 days to travel this route one way. On average, how many miles did the butterfly travel per day?

- A. 0.057
- B. 0.729
- C. 17.5
- D. 24

Each rock in a collection of 70 rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	33
sedimentary	27

If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

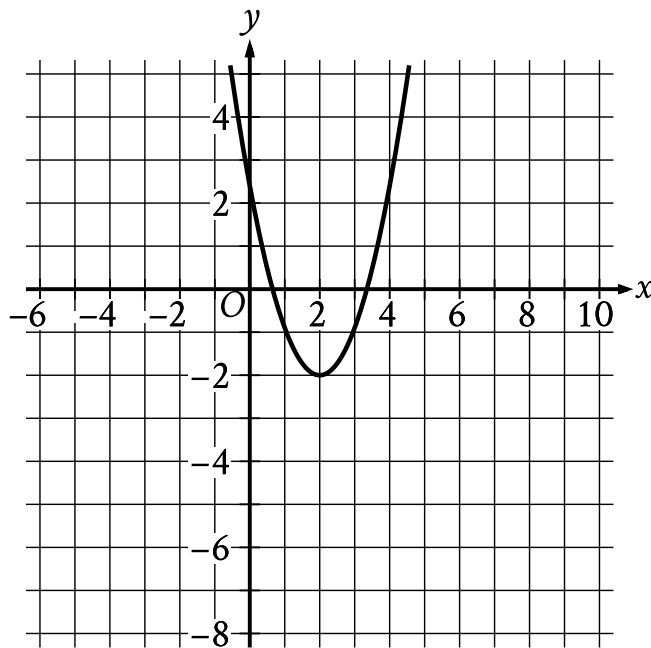
- A. $\frac{10}{27}$
- B. $\frac{10}{33}$
- C. $\frac{10}{60}$
- D. $\frac{10}{70}$

A company designs and makes handbags. To estimate the mean weight of the handbags made by the company on a particular day, a sample of the handbags made by the company on that day was selected at random. Based on the sample, it is estimated that the mean weight of all handbags made by the company on that day is 27.8 ounces oz, with an associated margin of error of 0.02 oz. Based on this estimate and associated margin of error, which of the following is the most plausible conclusion?

- A. The mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.
- B. The actual weights of all handbags made by the company on that day are between 27.78 oz and 27.82 oz.
- C. The actual weights of all handbags from the sample are between 27.78 oz and 27.82 oz.
- D. The mean weight of all handbags made by the company on that day is 27.8 oz.

Which expression is equivalent to $12x + 27$?

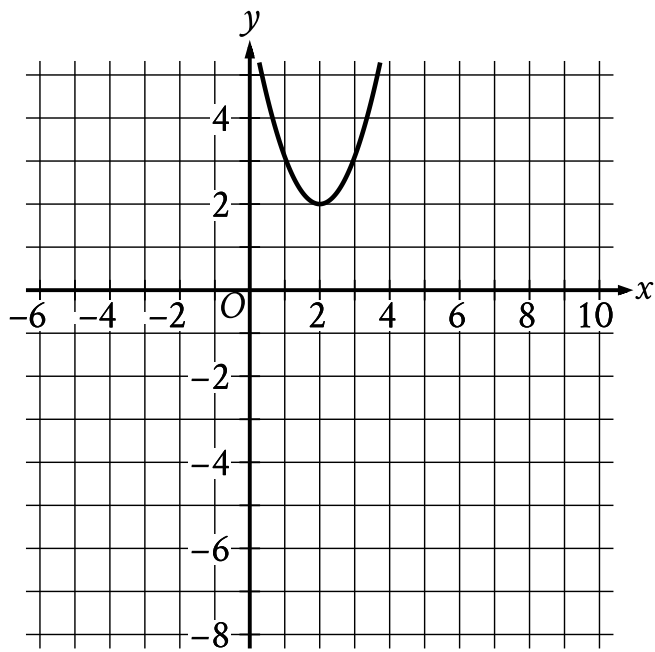
- A. $129x + 1$
- B. $2712x + 1$
- C. $34x + 9$
- D. $39x + 24$



- The parabola opens upward.
- The vertex is at the point (2 comma negative 2).
- The parabola passes through the following points:
 - (1 comma negative nine tenths)
 - (2 comma negative 2)
 - (3 comma negative nine tenths)

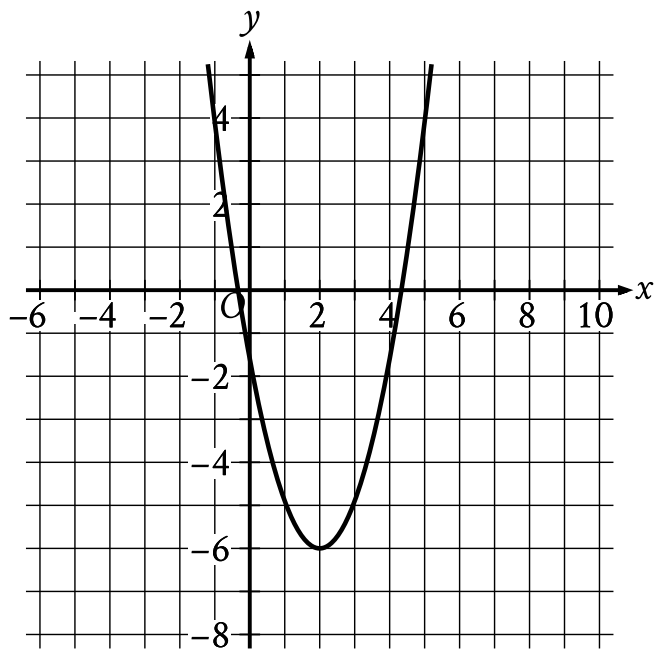
The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

A.



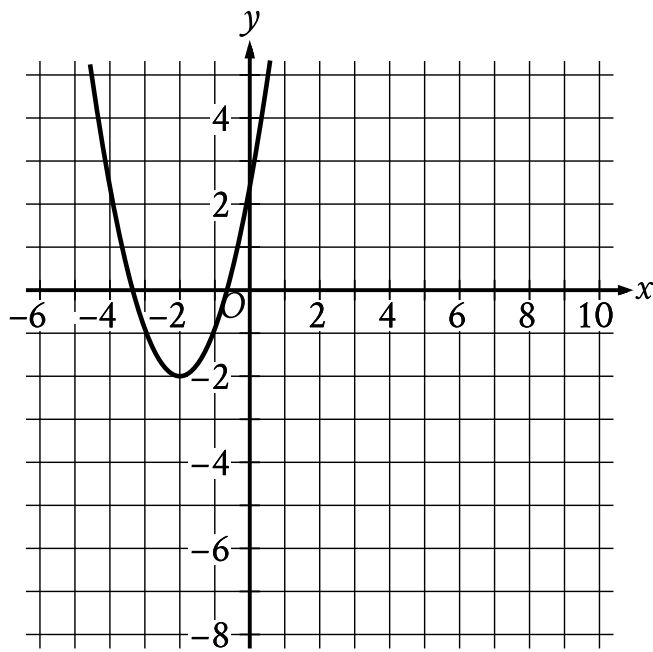
- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
 - (1 comma $\frac{31}{10}$)
 - (2 comma 2)
 - (3 comma $\frac{31}{10}$)

B.



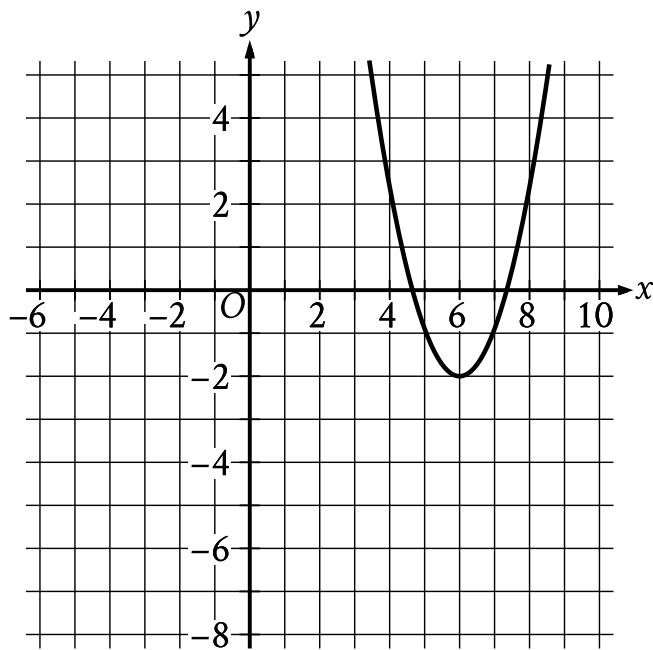
- The parabola opens upward.
- The vertex is at the point (2 comma negative 6).
- The parabola passes through the following points:
 - (1 comma negative $\frac{49}{10}$)
 - (2 comma negative 6)
 - (3 comma negative $\frac{49}{10}$)

C.



- The parabola opens upward.
- The vertex is at the point (negative 2 comma negative 2).
- The parabola passes through the following points:
 - (negative 3 comma negative nine tenths)
 - (negative 2 comma negative 2)
 - (negative 1 comma negative nine tenths)

D.



- The parabola opens upward.
- The vertex is at the point (6 comma negative 2).
- The parabola passes through the following points:
 - (5 comma negative nine tenths)
 - (6 comma negative 2)
 - (7 comma negative nine tenths)

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

A rectangle has a length of 64 inches and a width of 32 inches. What is the area, in square inches, of the rectangle?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.